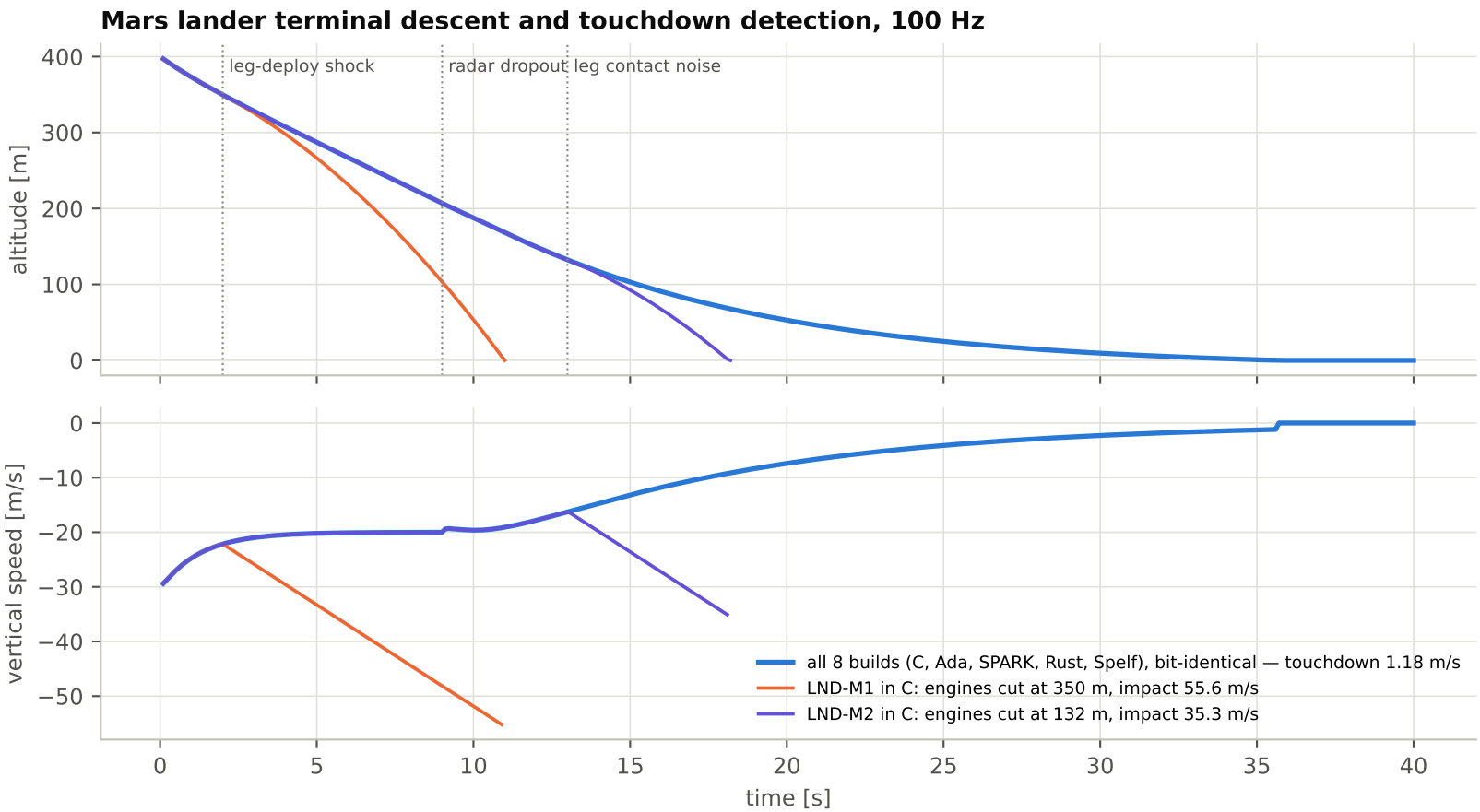
Mars lander terminal descent and touchdown detection, 100 Hz
leg-deploy shock
radar dropout
leg contact noise
altitude [m]
400
300
200
100
0
vertical speed [m/s]
0
−10
−20
−30
−40
−50
0
5
10
15
20
25
30
35
40
time [s]
all 8 builds (C, Ada, SPARK, Rust, Spelf), bit-identical — touchdown 1.18 m/s
LND-M1 in C: engines cut at 350 m, impact 55.6 m/s
LND-M2 in C: engines cut at 132 m, impact 35.3 m/s